INFOSYS SYSTEM ENGINEER

SECTION 1 — REASONING ABILITY TEST

15 Questions | 15 Marks | 25 Minutes

1. In a family, Arun is the son of Meena. Meena is the sister of Ravi. How is Ravi related to Arun?

A) Maternal uncle
B) Brother
C) Cousin
D) Father

2. Six people A, B, C, D, E and F sit around a circular table facing the centre. A sits between B and C. D sits opposite B. F sits immediately to the left of E. Who sits opposite A?

A) C
B) D
C) E
D) F

3. Statements: All engineers are graduates. Some graduates are artists. Conclusions: I. Some engineers are artists. II. All artists are graduates. Which conclusion(s) logically follow?

A) Only I
B) Only II
C) Both I and II
D) Neither I nor II

4. A standard die has opposite faces adding up to 7. If 6 is on the top face, which number is on the bottom face?

A) 1
B) 2
C) 5
D) 6

5. Five people P, Q, R, S and T stand in a row. P is immediately to the right of Q. R is immediately to the right of P. S is at one end and T is between S and Q. Who stands in the middle?

A) P
B) Q
C) R
D) T

6. A person walks 7 km north, turns right and walks 4 km, then turns right and walks 7 km. How far and in which direction is the person from the starting point?

A) 4 km east
B) 4 km west
C) 7 km east
D) 11 km south

7. Find the next letter in the series: A, D, G, J, M, __.

A) N
B) O
C) P
D) Q

8. Choose the odd one out.

A) Triangle
B) Square
C) Circle
D) Rectangle

9. If 15 August 1947 was a Friday, what day of the week was 26 January 2020?

A) Sunday
B) Monday
C) Saturday
D) Friday

10. Find the next number: 2, 5, 10, 17, 26, __.

A) 35
B) 36
C) 37
D) 38

11. John is twice as old as Mary. Five years ago, the sum of their ages was 30. What are their present ages?

A) 20 and 10
B) 22 and 11
C) 25 and 12.5
D) 24 and 12

12. At a meeting, every person shakes hands exactly once with every other person. If 10 people attend, how many handshakes take place?

A) 40
B) 45
C) 50
D) 55

13. If the day after tomorrow is Thursday, what day was yesterday?

A) Sunday
B) Monday
C) Tuesday
D) Wednesday

14. Find the next number: 3, 8, 15, 24, 35, __.

A) 46
B) 47
C) 48
D) 49

15. Complete the analogy: 5 : 25 :: 7 : ?

A) 35

B) 42
C) 49
D) 56

SECTION 2 — TECHNICAL ABILITY TEST (QUANTITATIVE APTITUDE)

10 Questions | 10 Marks | 35 Minutes

1. Find the 8th term of the arithmetic sequence 8, 14, 20, 26, ...

A) 50
B) 52
C) 54
D) 56

2. A number is increased by 20% and then decreased by 20%. What is the net percentage change?

A) 4% increase
B) 4% decrease
C) No change
D) 8% decrease

3. The average of five numbers is 24. Four of the numbers are 16, 21, 27 and 30. Find the fifth number.

A) 24
B) 25
C) 26
D) 28

4. A and B invest money in the ratio 3:5. If the total profit is ₹6400, what is B's share?

A) ₹2400
B) ₹3200
C) ₹4000
D) ₹4800

5. An article is bought for ₹800 and sold for ₹920. What is the profit percentage?

A) 12%
B) 15%
C) 18%
D) 20%

6. A car travels 180 km at 60 km/h. How much time does it take?

A) 2 hours
B) 2.5 hours
C) 3 hours
D) 3.5 hours

7. Pipe A fills a tank in 12 hours and Pipe B fills it in 18 hours. How long will they take together?

A) 6 hours 48 minutes
B) 7 hours 12 minutes
C) 8 hours
D) 9 hours

8. In how many ways can 5 distinct students be arranged in a row?

A) 25
B) 60
C) 100
D) 120

9. From 7 people, in how many ways can a committee of 3 be selected?

A) 21
B) 28
C) 35
D) 42

10. If SEND + MORE = MONEY, where each letter represents a distinct digit, what is the value of SEND?

A) 9567
B) 9568
C) 9569
D) 9576

SECTION 3 — VERBAL ABILITY TEST

20 Questions | 20 Marks | 20 Minutes

1. Choose the word closest in meaning to “lucid”.

A) Confusing
B) Clear
C) Hidden
D) Doubtful

2. Choose the word closest in meaning to “benevolent”.

A) Kind
B) Selfish
C) Cruel
D) Careless

3. Choose the word opposite in meaning to “frugal”.

A) Economical
B) Thrifty
C) Extravagant
D) Careful

4. Choose the word closest in meaning to “obstinate”.

A) Flexible
B) Stubborn
C) Helpful
D) Polite

5. Choose the word closest in meaning to “erudite”.

A) Learned
B) Uncertain
C) Ordinary
D) Silent

6. What does the idiom “break the ice” mean?

A) To end a friendship
B) To start a conversation comfortably
C) To create a problem
D) To refuse an invitation

7. What does “hit the sack” mean?

A) Go to sleep
B) Start working
C) Become angry
D) Leave a place permanently

8. What does “once in a blue moon” mean?

A) Very frequently
B) At night
C) Very rarely
D) Unexpectedly

9. What does “cost an arm and a leg” mean?

A) Be very expensive
B) Be dangerous
C) Be free
D) Take very little time

10. What does “beat around the bush” mean?

A) Speak directly
B) Avoid the main point
C) Work quickly
D) Argue loudly

11. Choose the correct option: He ____ to school when it started to rain.

A) walks
B) was walking
C) has walked
D) will walk

12. Choose the correct option: If I ____ you, I would apologize.

A) am
B) was
C) were
D) be

13. Choose the correct option: By this time tomorrow, I ____ my project.

A) finish
B) finished
C) will have finished
D) am finishing

14. Choose the correct option: She ____ in Paris since 2010.

A) lives

B) lived
C) has lived
D) is living

15. Choose the correct option: They would go outside ____ it was raining.

A) unless
B) because
C) although
D) so that

16. Identify the part containing the error: “Neither of the students were present in the class.”

A) Neither
B) of the students
C) were present
D) in the class

17. Identify the part containing the error: “She don’t like apples.”

A) She
B) don’t
C) like
D) apples

18. Identify the part containing the error: “I have visited London last year.”

A) I have
B) visited
C) London
D) last year

19. Identify the part containing the error: “Him and I are going to the store.”

A) Him and I
B) are going
C) to the
D) store

20. Identify the part containing the error: “It is the best book I have ever read it.”

A) It is
B) the best book
C) I have ever read
D) it

SECTION 4 — PSEUDO CODE TEST

5 Questions | 10 Marks | 10 Minutes

1. Consider the following pseudocode:

```
A = [3, 1, 4, 1]
FOR i = 0 TO 2
  SWAP A[i] AND A[i+1]
END FOR
PRINT A
```

What is the output?

A) [1, 4, 1, 3]

B) [1, 3, 1, 4]

C) [3, 1, 1, 4]

D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10

B) 20

C) 30

D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12

B) 16

C) 24

D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)

B) O(log N)

C) $O(N^2)$

D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

```
  IF x MOD 2 == 0
    count = count + 1
  END IF
END FOR
PRINT count
```

What is printed?

A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:

```
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E
```

What is the minimum number of moves required to reach E from S?

A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

```
2  5  9
4  8  15
6  11  ?
```

Choose the number that logically completes the pattern.

A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

Grid:

S(1) 3 5 8

2 4 6 9

1 2 7 10

0 1 8 E(11)

Which of the following gives a valid path from S to E?

A) 1→2→4→6→9→10→11

B) 1→3→4→2→7→10→11

C) 1→2→1→7→8→11

D) 1→5→6→7→8→11

4. Missing-Cell Grid Puzzle:

Each row and each column follows the same numerical rule.

3 6 9

5 10 15

7 14 ?

What number should replace ?

A) 18

B) 20

C) 21

D) 24

SECTION 6 — ENGLISH GRAMMAR TEST

5 Questions | 10 Marks | 10 Minutes

1. Nobody in the class ____ finished the assignment on time.

A) have

B) has

C) having

D) were

2. Either the coach or the players ____ going to attend the meeting.

A) is

B) was

C) are

D) has

3. I want him ____ to the party.

A) come

B) coming

C) to come

D) came

4. The cookies ____ baked by Sarah tasted delicious.

A) was

B) were

C) is

D) has

5. Each of the players ____ received a certificate.

A) have

B) has

C) are

D) were

SECTION 7 — ENGLISH WRITTEN TEST

1 Question | Evaluated | 10 Minutes

1. Write an essay of approximately 150–200 words on the following topic:

“Effective communication is one of the most important skills for success in a professional workplace.”